Table 1: Oracle threshold variation results for failure rate for Judge-based effectiveness fitness f_{dim} and efficiency fitness function $f_{efficient}$. Structured fitness f_{eff} threshold is kept 0.75 (balancing borderline fail vs. complete pass). Statistically significant differences using Wilcoxon signed-rank test (p-value below 0.05) comparing (LUNAR/S vs. SENSEI) are highlighted in bold. If results are not statistically significant, neither approach has superior performance. Suffix I represents open-source SUT and suffix II represents industrial SUT. Effect sizes are Large. Note: Less constraining higher thresholds (i.e., 0.9, 1.0) partially lead to large failure spaces making all approaches fail similarly.

(a) Carcontrol-I

LLM	Approach	0.5	0.6	0.7	0.8	0.9	1.0
gpt-5-chat	LUNAR	**0.274**	**0.312**	**0.313**	**0.313**	**0.313**	**0.313**
	LUNAR_S	**0.095**	**0.124**	**0.124**	**0.124**	**0.124**	**0.124**
	SENSEI	0.005	0.042	0.062	0.062	0.096	0.096
gpt-4o	LUNAR	**0.334**	**0.470**	**0.470**	**0.470**	**0.470**	**0.470**
	LUNAR_S	0.112	0.231	0.233	0.233	0.233	0.233
	SENSEI	0.000	0.117	0.140	0.140	0.162	0.162

(b) Navi-I

LLM	Approach	0.5	0.6	0.7	0.8	0.9	1.0
gpt-5-chat	LUNAR	**0.574**	**0.591**	**0.597**	**0.597**	**0.597**	**0.597**
	LUNAR_S	**0.198**	**0.255**	0.264	0.264	0.264	0.264
	SENSEI	0.013	0.051	0.339	0.339	0.390	0.390
gpt-4o	LUNAR	**0.588**	**0.612**	**0.631**	**0.631**	**0.631**	**0.631**
	LUNAR_S	**0.283**	**0.330**	**0.341**	**0.341**	**0.341**	**0.341**
	SENSEI	0.028	0.051	0.181	0.181	0.238	0.238

(c) Carcontrol-II

LLM	Approach	0.5	0.6	0.7	0.8	0.9	1.0
gpt-5-chat	LUNAR	0.002	0.140	0.140	0.140	0.140	0.140
	LUNAR_S	0.003	**0.171**	**0.171**	**0.171**	**0.171**	**0.171**
	SENSEI	0.000	0.037	0.075	0.075	0.088	0.088
gpt-4o	LUNAR	**0.005**	**0.115**	**0.115**	**0.115**	**0.115**	**0.115**
	LUNAR_S	**0.007**	**0.073**	**0.073**	**0.073**	**0.073**	**0.073**
	SENSEI	0.000	0.000	0.000	0.000	0.004	0.004

(d) Navi-II

LLM	Approach	0.5	0.6	0.7	0.8	0.9	1.0
gpt-5-chat	LUNAR	**0.221**	**0.252**	**0.261**	**0.261**	0.261	0.261
	LUNAR_S	**0.185**	**0.236**	**0.237**	**0.237**	0.237	0.237
	SENSEI	0.002	0.047	0.119	0.119	0.578	0.578
gpt-4o	LUNAR	**0.225**	**0.305**	**0.305**	**0.305**	0.305	0.305
	LUNAR_S	**0.174**	**0.278**	**0.278**	**0.278**	0.278	0.278
	SENSEI	0.002	0.106	0.231	0.231	0.524	0.524